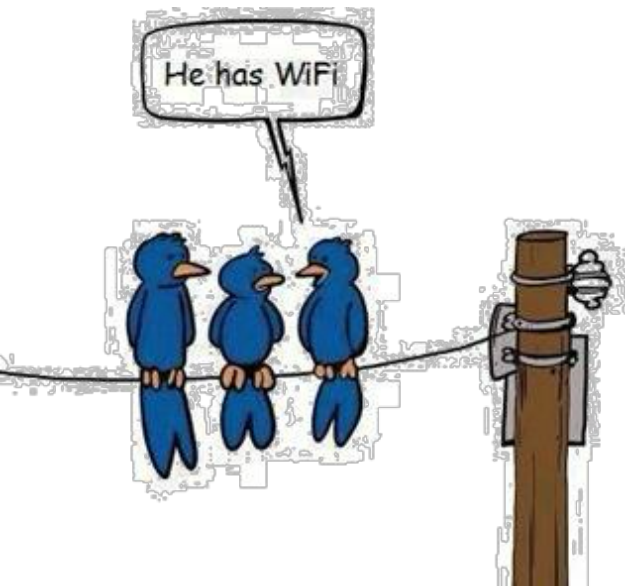


ECSE 4560/ 6560

Modern Communication Systems

Lecture 20: MIMO Review and Wireless Sensing Part 1

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Sensing**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Project mid-term report due **Monday 11/10**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. Late due (Wed 11/13)
 - HW6 (MIMO code) due **Fri 11/14**
 - HW7 (Sensing Theory) due **Fri 11/21** – covered in Exam 2
 - Exam 2: **Friday Nov 21** - Anyone needs makeup exam?

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**
- Sensing Readings
 - Xiong, Jie, and Kyle Jamieson. "ArrayTrack: A Fine-Grained indoor location system." *NSDI 2013*
 - 2024 SIGMOBILE Test-of-Time Award 

Today's Class

- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- Intro-Wireless Sensing
 - Localization Techniques

MIMO Gains

Multiplexing Gain:

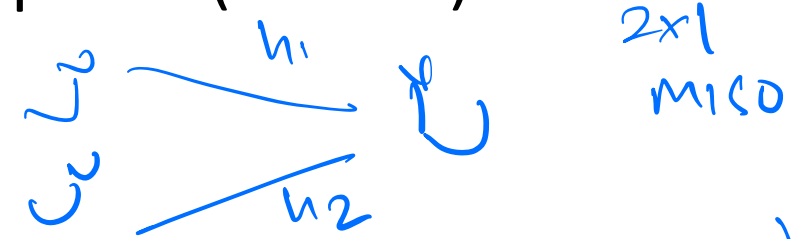
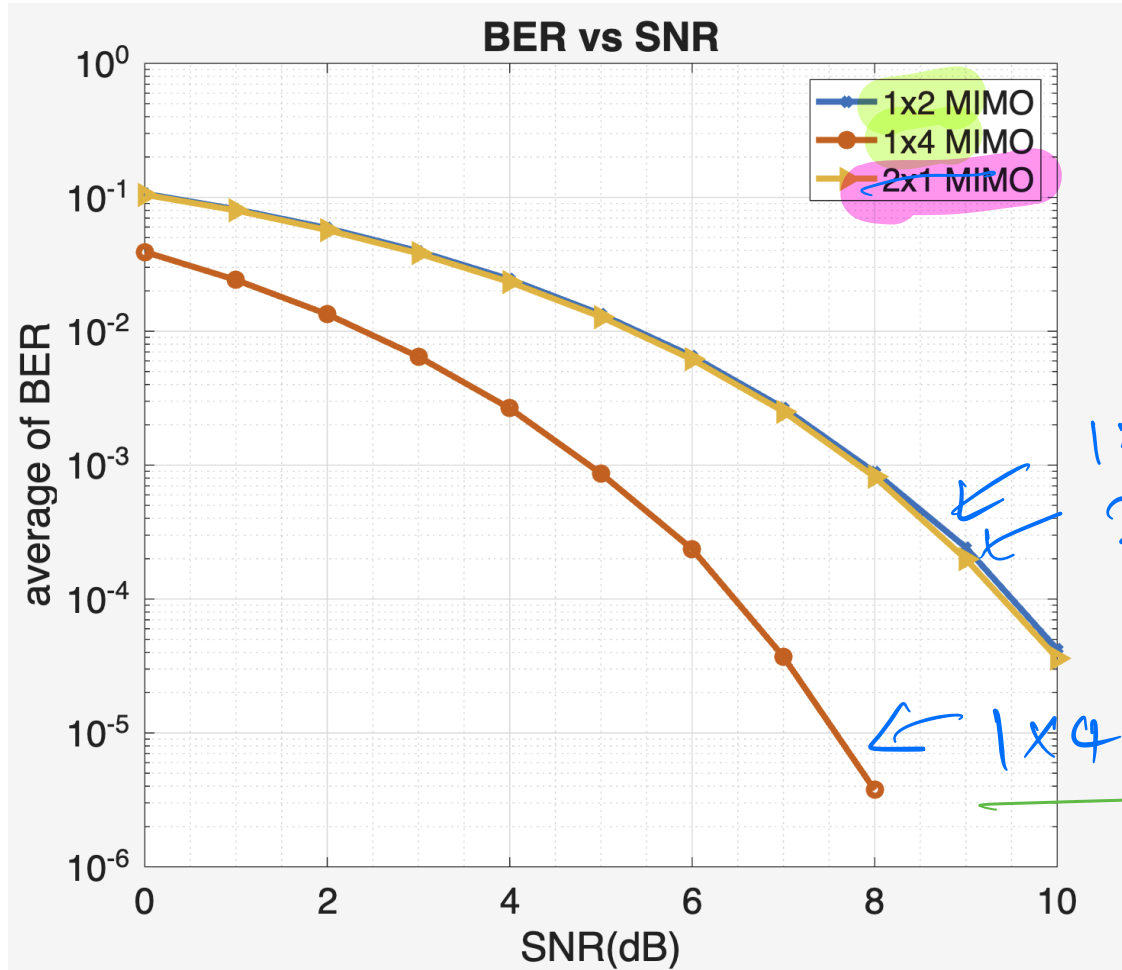
- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow \times N$ more packets
- Data Rate: $\propto N \log(SNR/N)$

Diversity Gain:

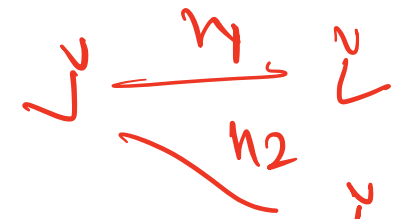
- Increase SNR of the received packets
- $N \times 1$ or $1 \times N$ MIMO $\rightarrow \times \log N$ rate
- $N \times N$ MIMO $\rightarrow \times \log N^2$ data rate
- Data Rate $\propto \log(SNR \times N^2)$

$\uparrow$
(SISO
1x1 SNR)

OFDM-MIMO Diversity Techniques (HW6)



SNR $\propto (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2}$
 (assume CSIT)



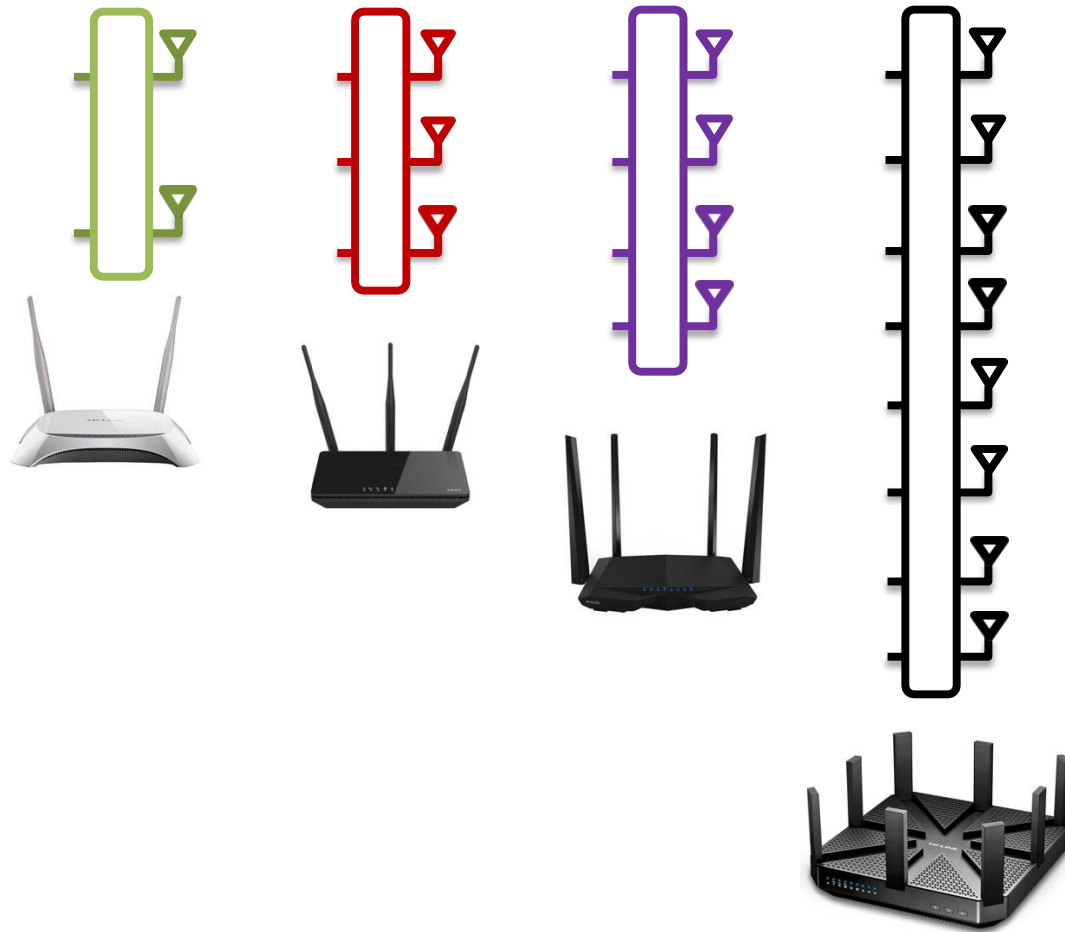
SNR $\propto (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2}$



SNR $\propto (|h_1|^2 + |h_2|^2 + |h_3|^2 + |h_4|^2) \frac{P}{\sigma^2}$

MIMO: Multiple TX-RX streams

WiFi Access Points 2 to 8 MIMO

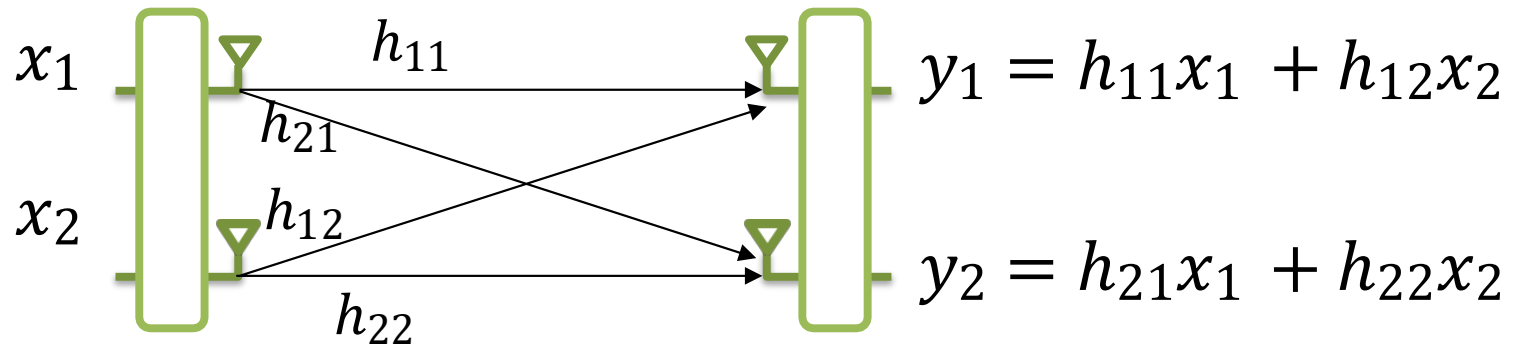


WiFi Devices 1 to 2 MIMO



- Power
- Form Factor & Antenna Separation
- Most mobile phones support 2 MIMO

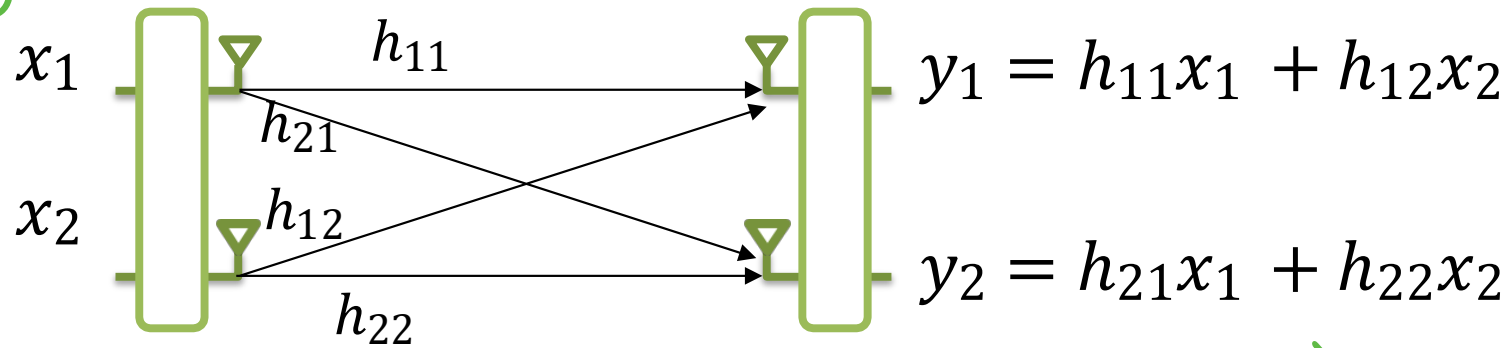
MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Zero-forcing-based MIMO



$$y = \underbrace{Hx + n}$$

$$\hat{x} = (H^H H)^{-1} H^H y$$

$$= (H^H H)^{-1} H^H (Hx + n)$$

$$= \left(\cancel{(H^H H)^{-1} H^H} \right) x + (H^H H)^{-1} H^H n$$

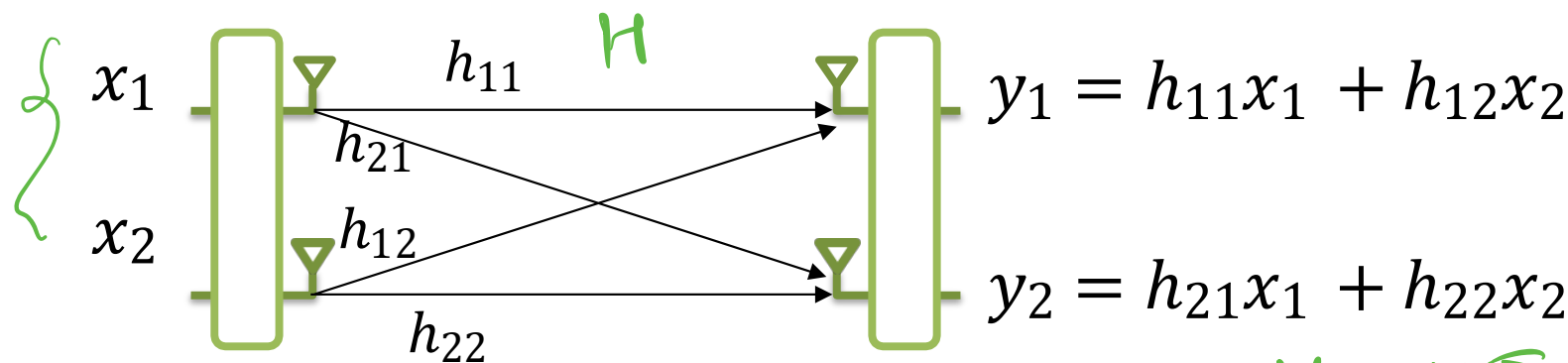
completely removes interference from multiple transmit streams (x_1 & x_2)

But compromises SNR per-stream

$\min(N_{Tx}, N_{Rx})$

SVD-based MIMO

CSIT $H_{2 \times 2}$



$$y_1 = h_{11}x_1 + h_{12}x_2$$

$$y_2 = h_{21}x_1 + h_{22}x_2$$

$$H = U \Sigma V^H$$

↑
singular values

⇒ multiple parallel
SISO channels
⇒ each channel have
diff SNR which is
prop to σ_1, σ_2

$$H = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{u_1} & \underbrace{u_{r2}}_{u_2} & \dots & \underbrace{u_{rt}}_{u_t} \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\Sigma}$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{V^H}$$

$$= U \Sigma V^H$$

$$y = Hx + n$$

$$= U \Sigma V^H x + n$$

$$y = Hx + n$$

$$y = U \Sigma V^H \tilde{x} + n$$

$$U^H y = \cancel{U^H U} \Sigma \cancel{V^H V} \tilde{x}$$

ofdm waveform

$$\tilde{y} = U^H y = \Sigma \begin{pmatrix} \tilde{x} \\ 0 \end{pmatrix}$$

$$\tilde{y} = \Sigma \tilde{x}$$

$$\sigma_1 > \sigma_2$$

$$y_1 = \sigma_1 x_1$$

$$H y_2 = \sigma_2 x_2$$

$$H = U \Sigma V$$

$$H = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} v_1^H \\ v_2^H \end{bmatrix} \quad H = \begin{bmatrix} \bar{v}_2 & \bar{v}_1 \end{bmatrix} \begin{bmatrix} \sigma_2 & 0 \\ 0 & \sigma_1 \end{bmatrix} \begin{bmatrix} v_2^H \\ v_1^H \end{bmatrix}$$

not invertible

$$P =$$

$$P_{inv}(H) = \underbrace{(H^H H)^{-1}}_{3 \times 3} H^H \quad \begin{matrix} 3 \times 2 & 2 \times 3 & 3 \times 2 \\ & 3 \times 3 & \\ & & 3 \times 2 \end{matrix}$$

$$\begin{aligned} y &= Hx \\ P^T y &= P^T Hx \\ 3 \times 2, 2 \times 3 & \\ 3 \times 3 & \end{aligned}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_1 \end{bmatrix}$$

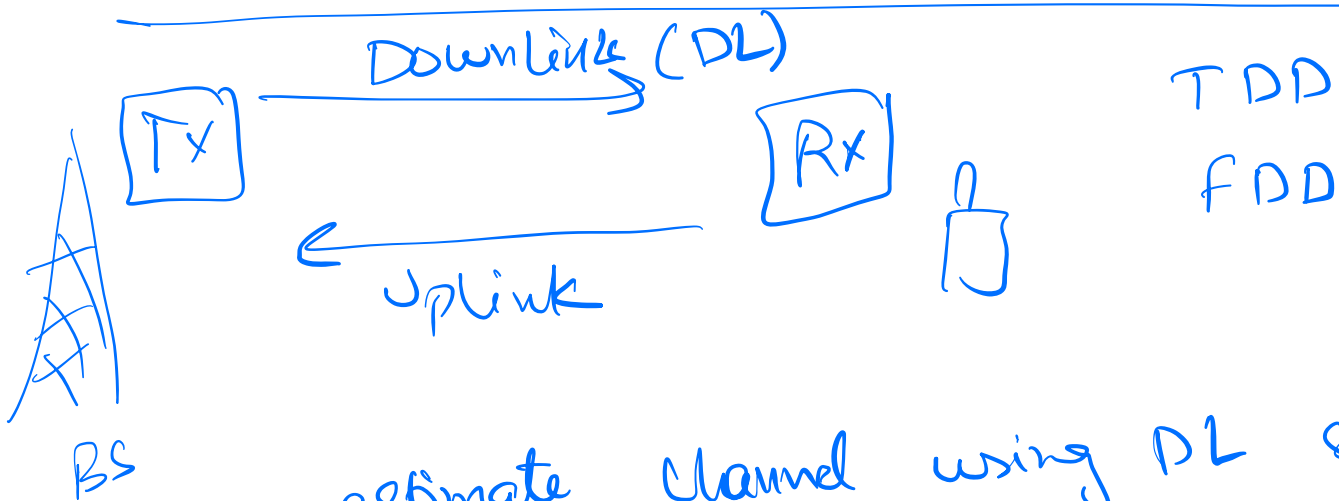
$$H \quad \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix}$$

total # non zero-eigen values

of $N \times M$ matrix H

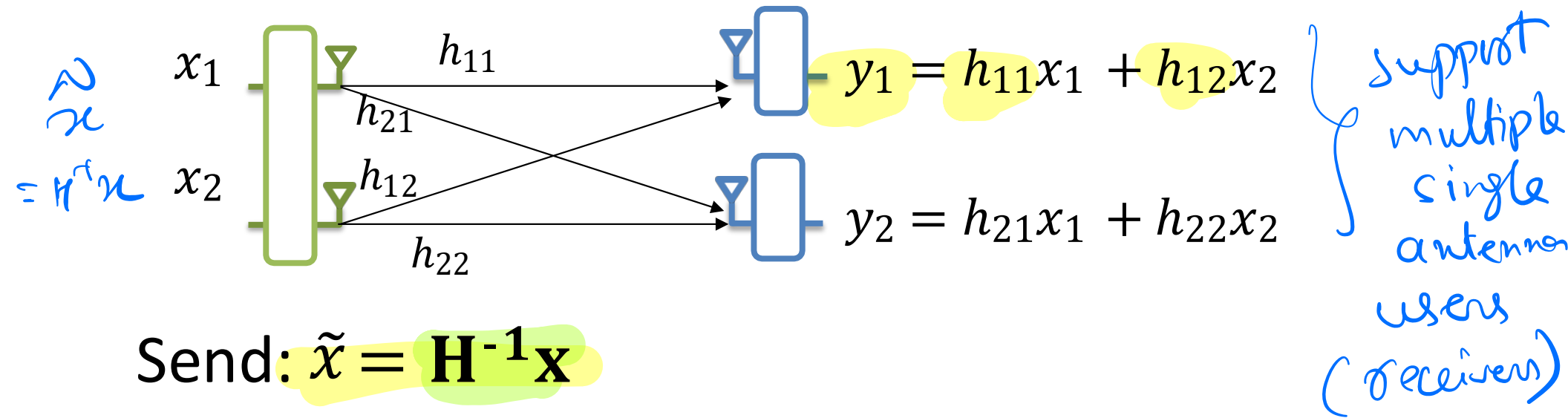
is at most

$\min(N, M)$



estimate channel using DL slot
feedback using next UL slot
— control channel —

MU-MIMO Precoding



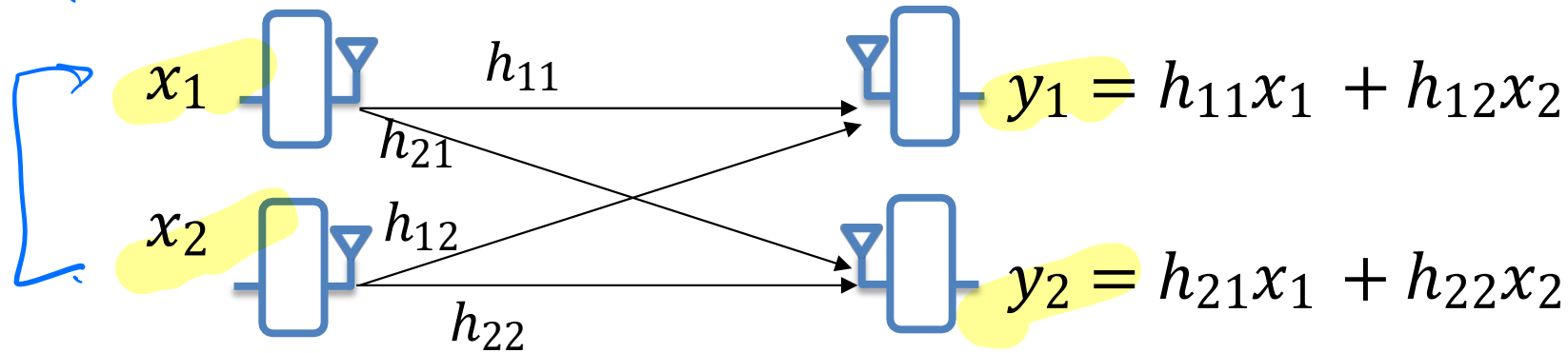
Receive: $\mathbf{y} = \mathbf{H} \tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H} \mathbf{H}^{-1} \mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Also known as Beamforming or Zero-Forcing

Requires Channel Feedback!

Synchronize Tx with cable

Distributed MIMO



Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Does not work. Transmitters do not have access to each other's signals

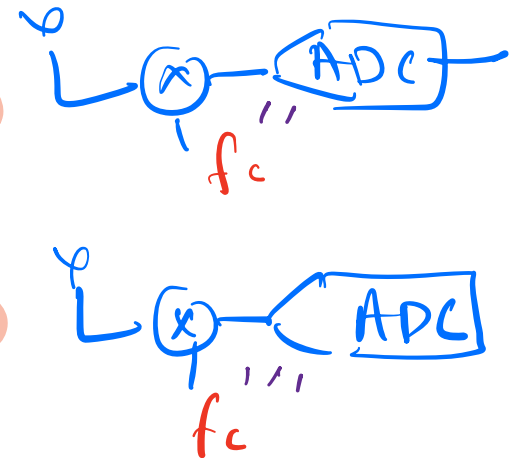
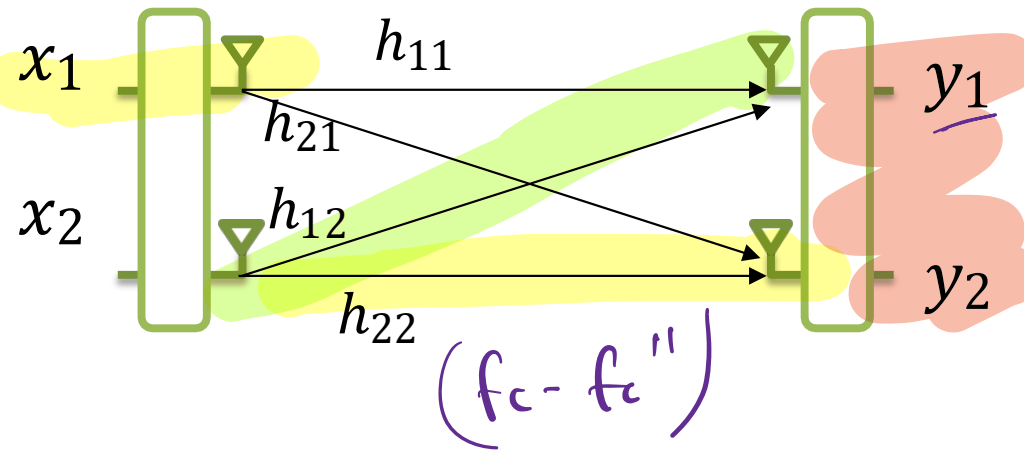
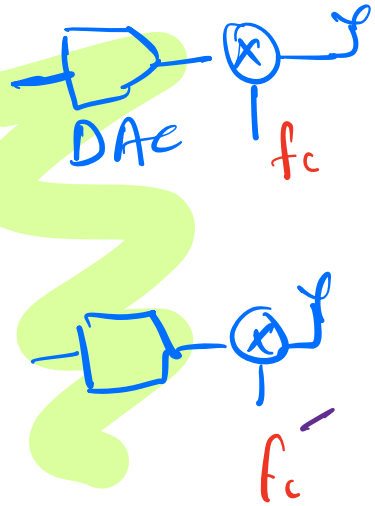
False: what if transmitters are APs connected over Ethernet?

Today's Class

- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- Intro-Wireless Sensing
 - Localization Techniques

Clock Synchronization in MIMO

$$f_c'' = f_c'''$$



$$y_1 = h_{11}x_1 e^{-j2\pi \Delta f_{11} t} + h_{12}x_2 e^{-j2\pi \Delta f_{12} t} \quad f_c'' - f_c'$$

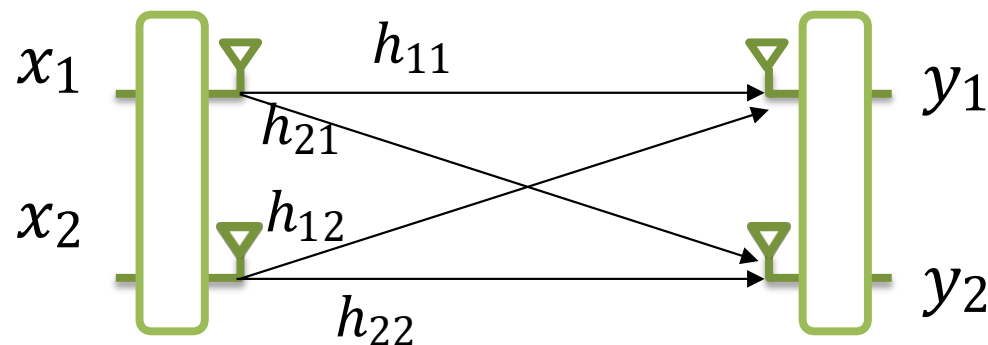
$$y_2 = h_{21}x_1 e^{-j2\pi \Delta f_{21} t} + h_{22}x_2 e^{-j2\pi \Delta f_{22} t}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

$$y = Hx + n$$

$$f_c = f_c'$$

Clock Synchronization in MIMO



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

On chip MIMO: $\Delta f_{11} = \Delta f_{21} = \Delta f_{12} = \Delta f_{22} = \Delta f$

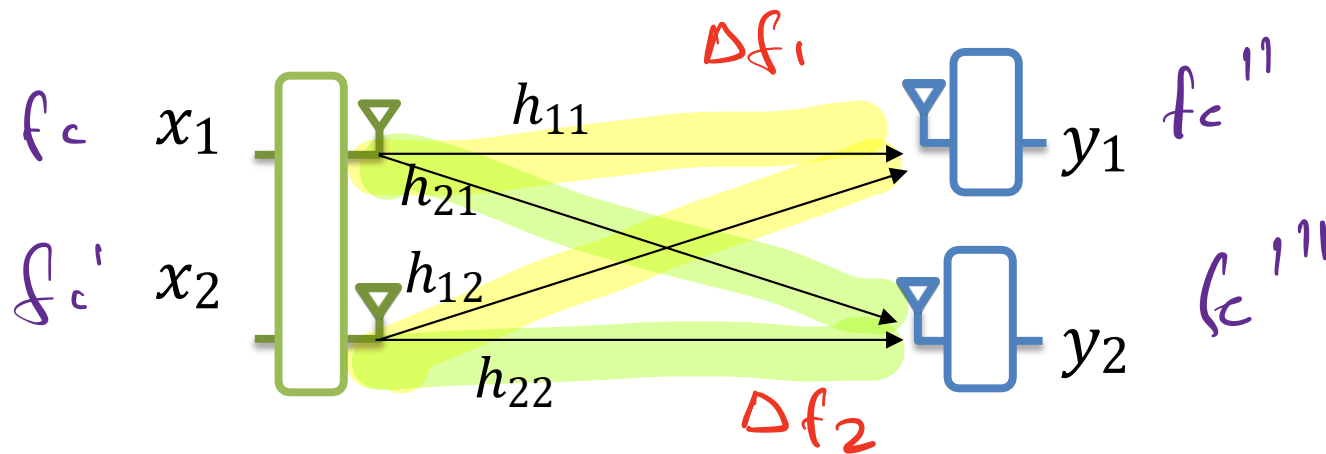
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} e^{-j2\pi\Delta f t} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Handwritten notes: $e^{-j2\pi\Delta f t}$ is circled in purple. An arrow points from the text "CFO => correct it" to the circled term. To the right, it says "similar to SISO" and "ofdm".

Clock Synchronization in MIMO

mumimo
 $\left[\frac{f_c''}{f_c'''} \right]$

$f_c = f_c'$

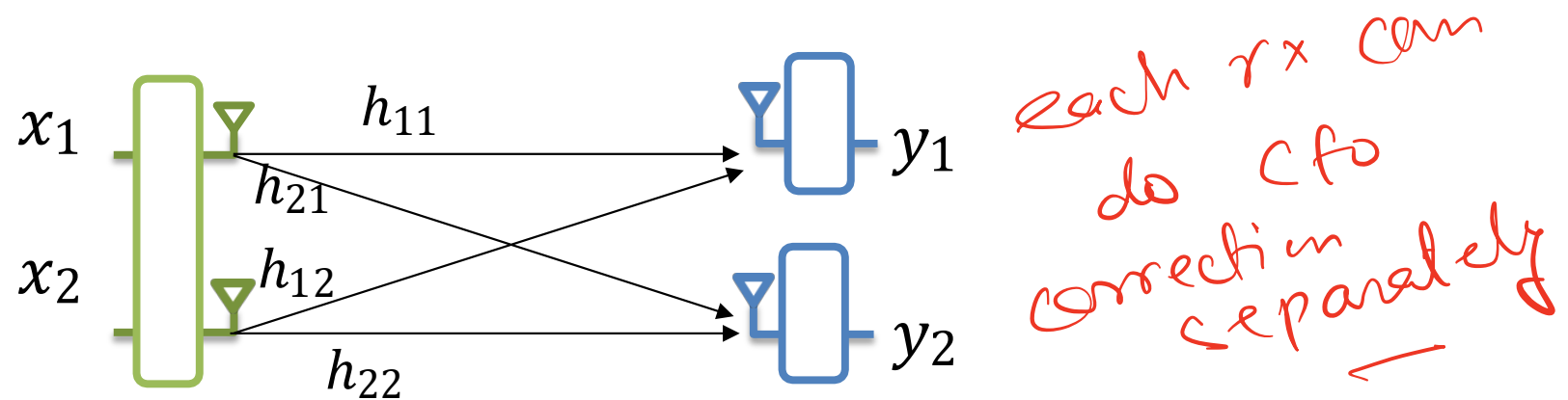


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_1t} & h_{12} e^{-j2\pi\Delta f_1t} \\ h_{21} e^{-j2\pi\Delta f_2t} & h_{22} e^{-j2\pi\Delta f_2t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

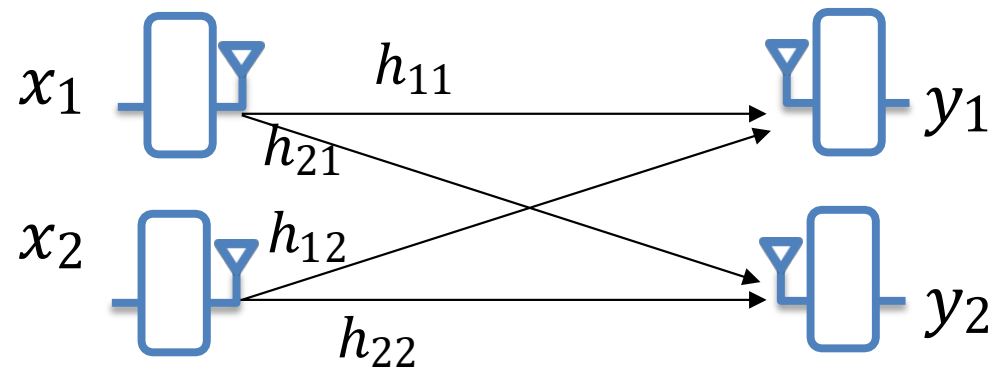


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e^{-j2\pi\Delta f_1 t} & 0 \\ 0 & e^{-j2\pi\Delta f_2 t} \end{bmatrix} \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Distributed MIMO: $\Delta f_{11} \neq \Delta f_{12} \neq \Delta f_{21} \neq \Delta f_{22}$

'mega mimo'

1 AC
interf. alignment
& cancellation

Today's Class

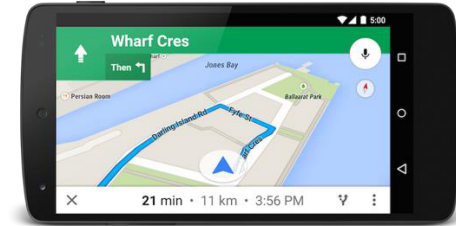
- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- **Intro-Wireless Sensing**
 - **Localization Techniques**

Wireless Localization / Positioning

The process of obtaining a human or object's location using wireless signals

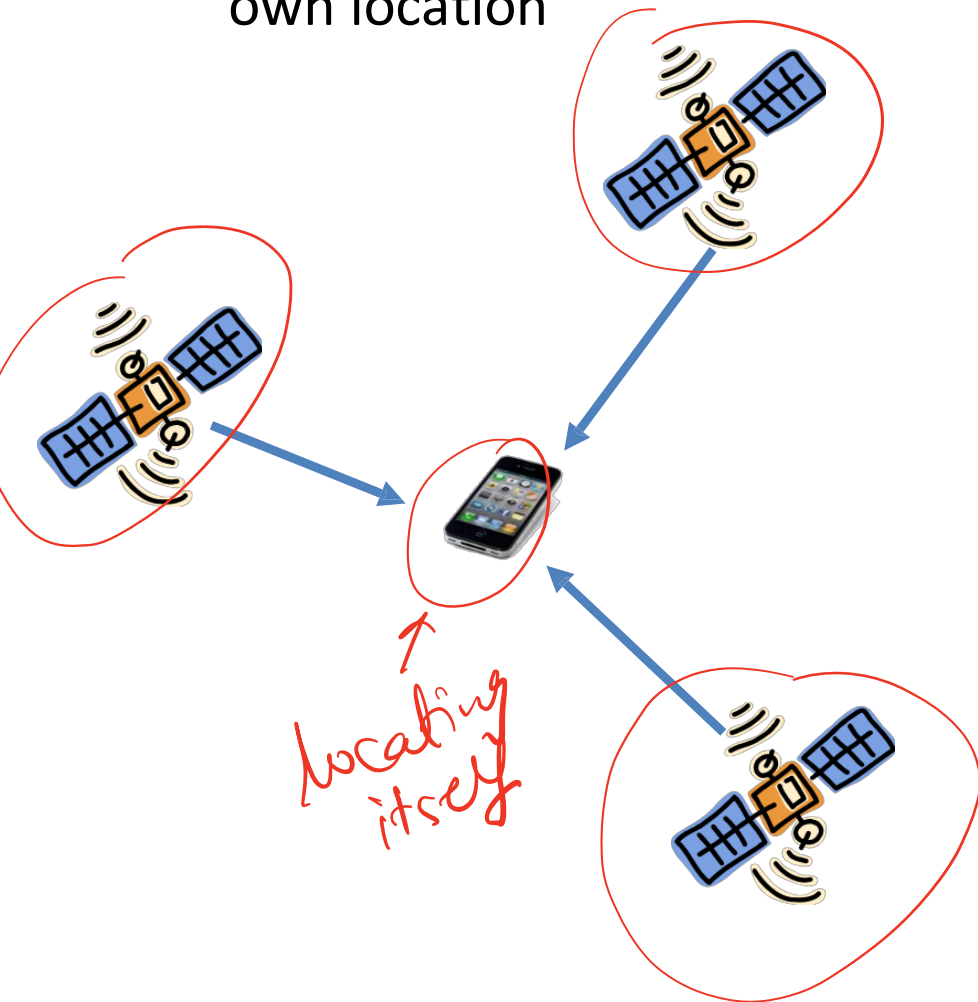
Applications:

- Navigation: outdoors (GPS) and indoors (e.g., museum)
- Location based services: Tagging, Reminder, Ads
- Virtual Reality and Motion Capture
- Gestures, writing in the air
- Behavioral Analytics (Health, activities, etc.)
- Locating misplaced items (keys)
- Location based security
- Delivery drones

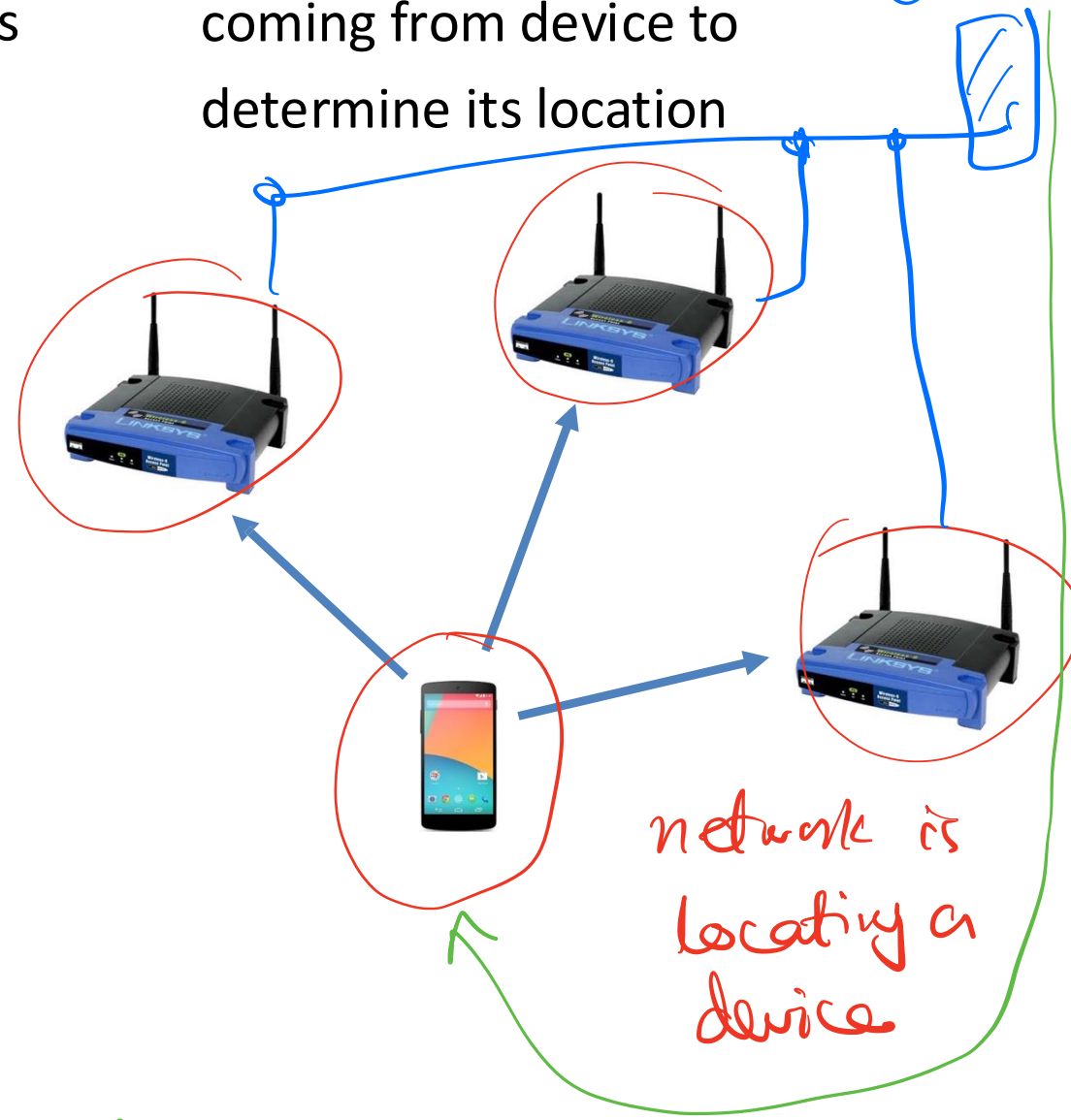


Wireless Localization Architecture.

- Device based: A device uses incoming signal from one or more “anchors” to determine its own location



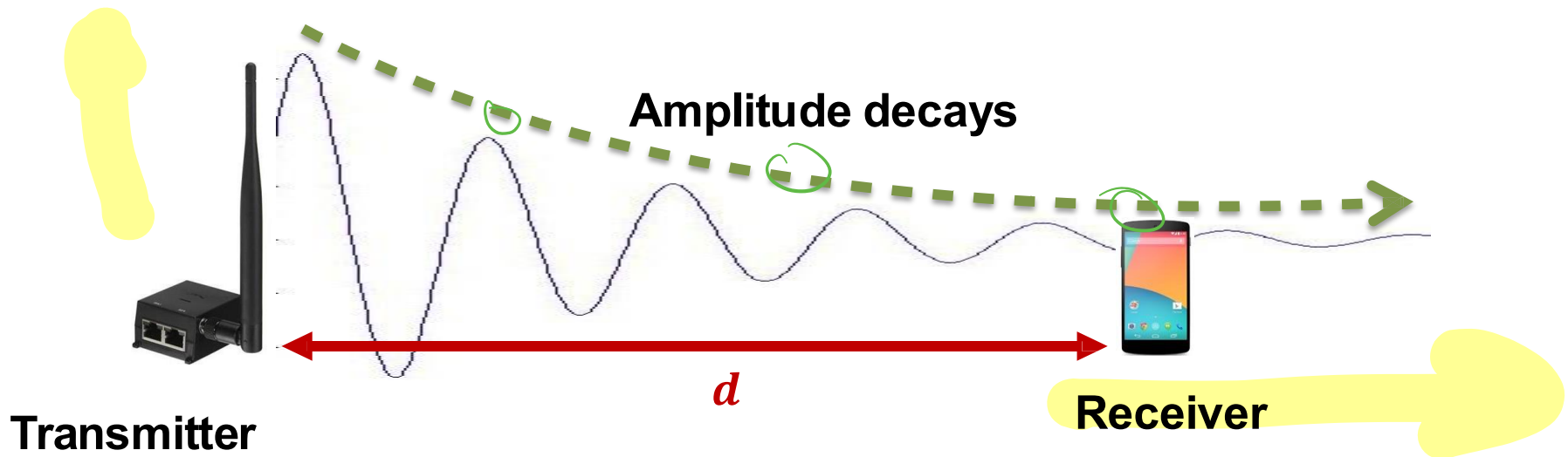
- Network based: Anchors (or Access points) use the signal coming from device to determine its location



Method 1: RSSI Based Localization

- Higher received power → Closer
- Lower received power → Farther

} estimate distance

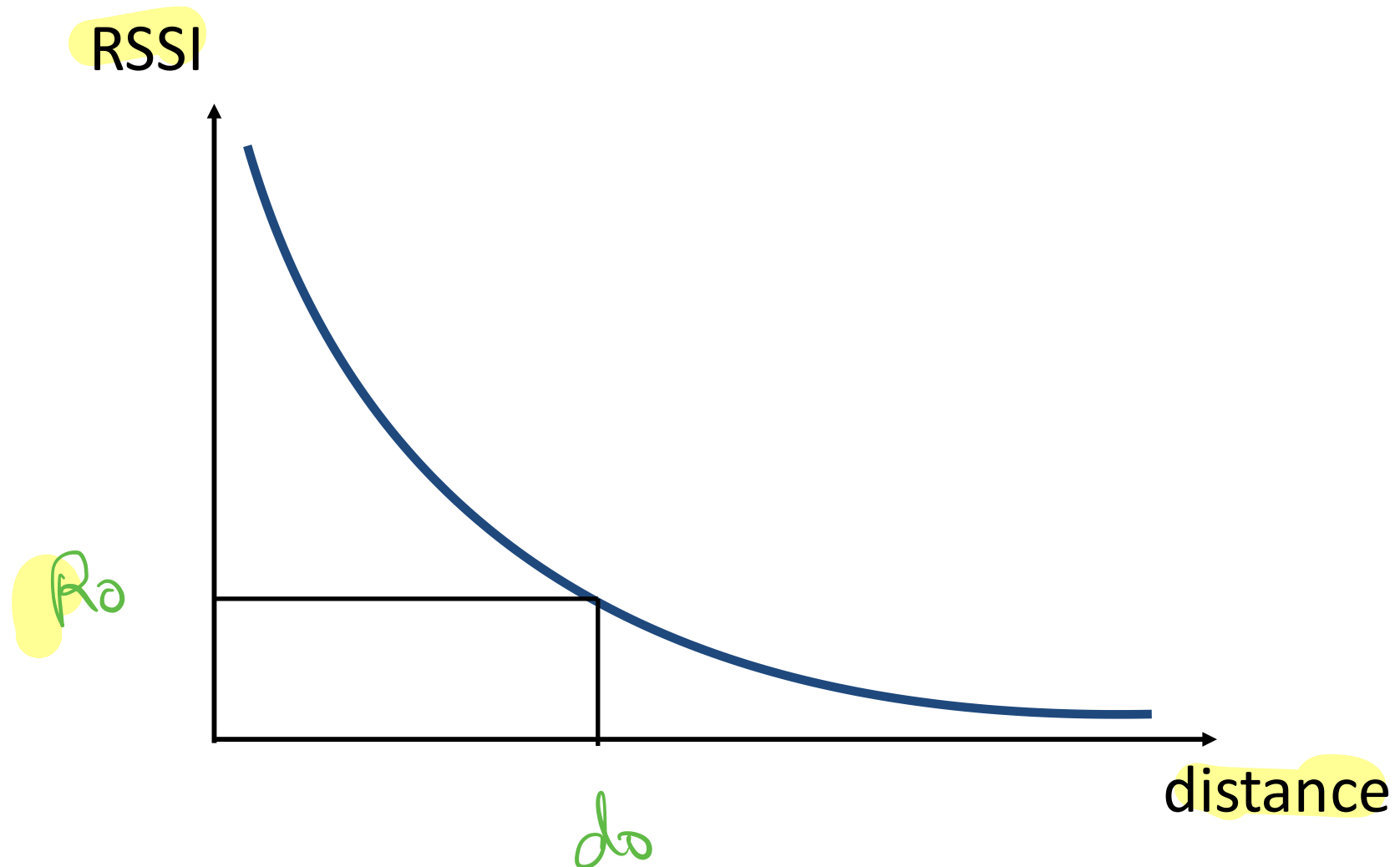


$$P_{Rx} = \frac{G_{Tx} G_{Rx} \lambda^2}{(4\pi d)^2} P_{Tx} \rightarrow RSSI \propto \frac{1}{d^2}$$

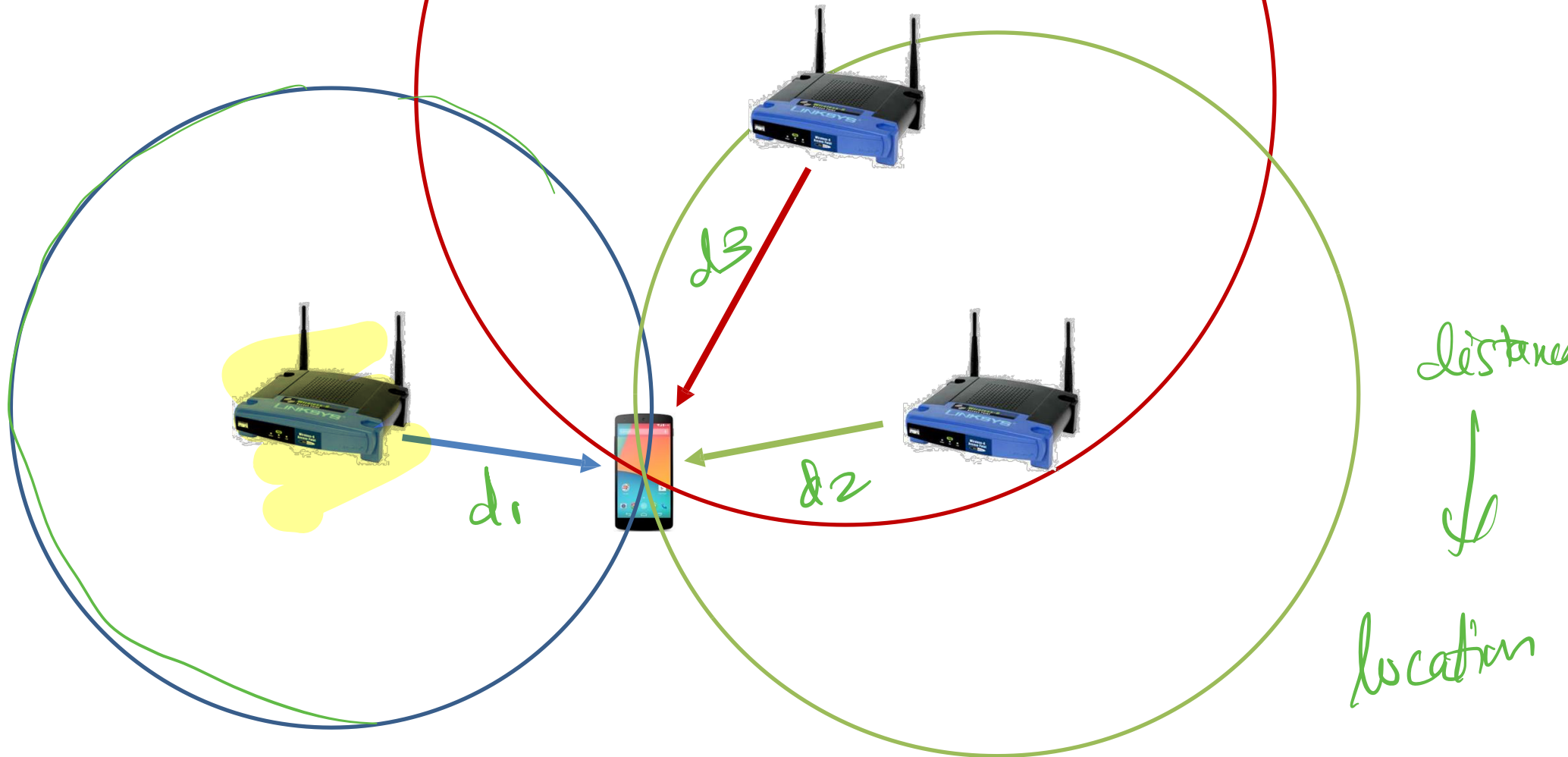
Use RSSI to estimate distance from APs!

Method 1: RSSI Based Localization

$$RSSI \propto \frac{1}{d^2}$$



Method 1: RSSI Based Localization

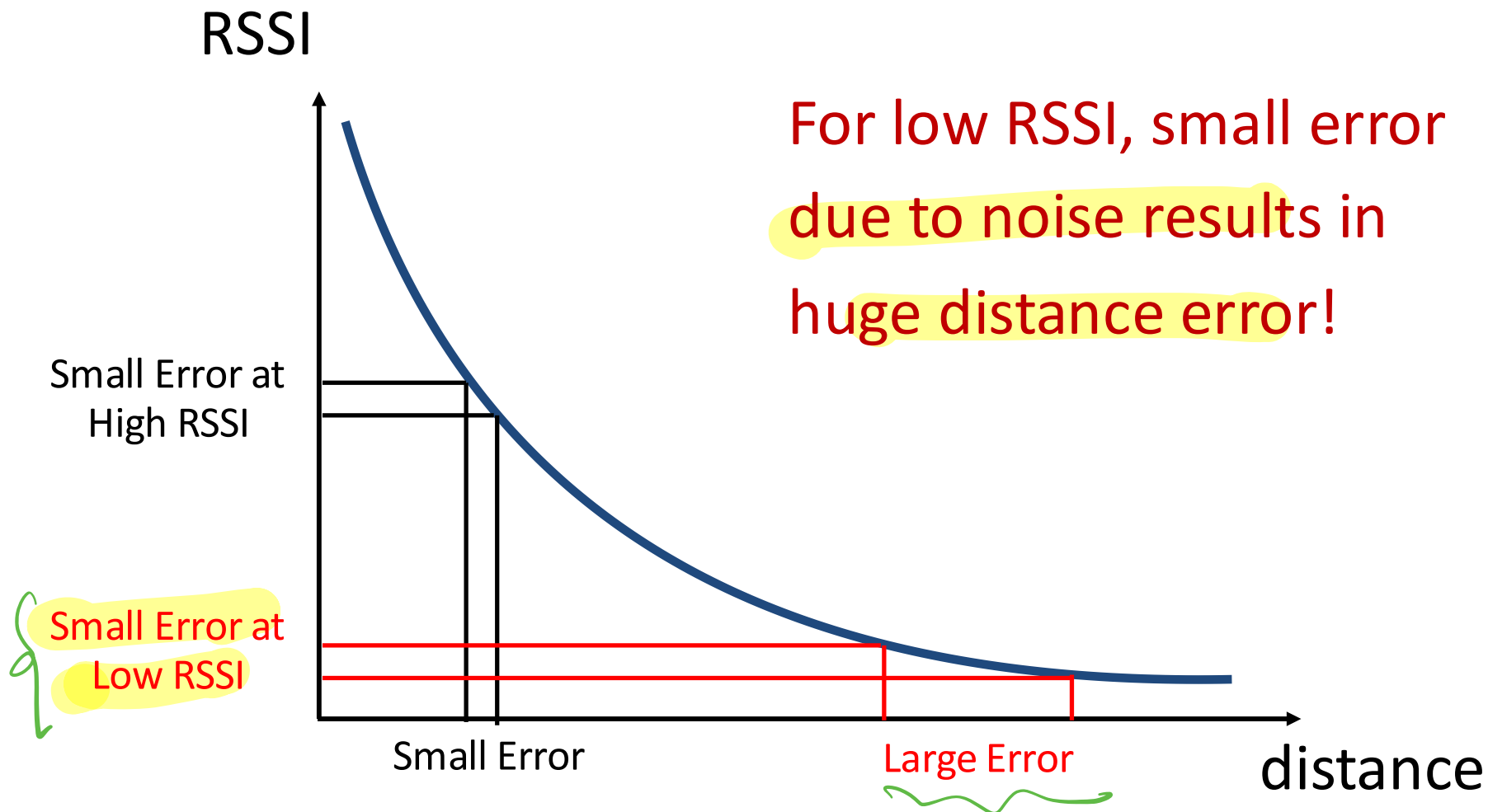


Trilateration $\Rightarrow$ need atleast 3 APs

Method 1: RSSI Based Localization

Pros: Very simple, no hardware modifications

Cons: Highly inaccurate!



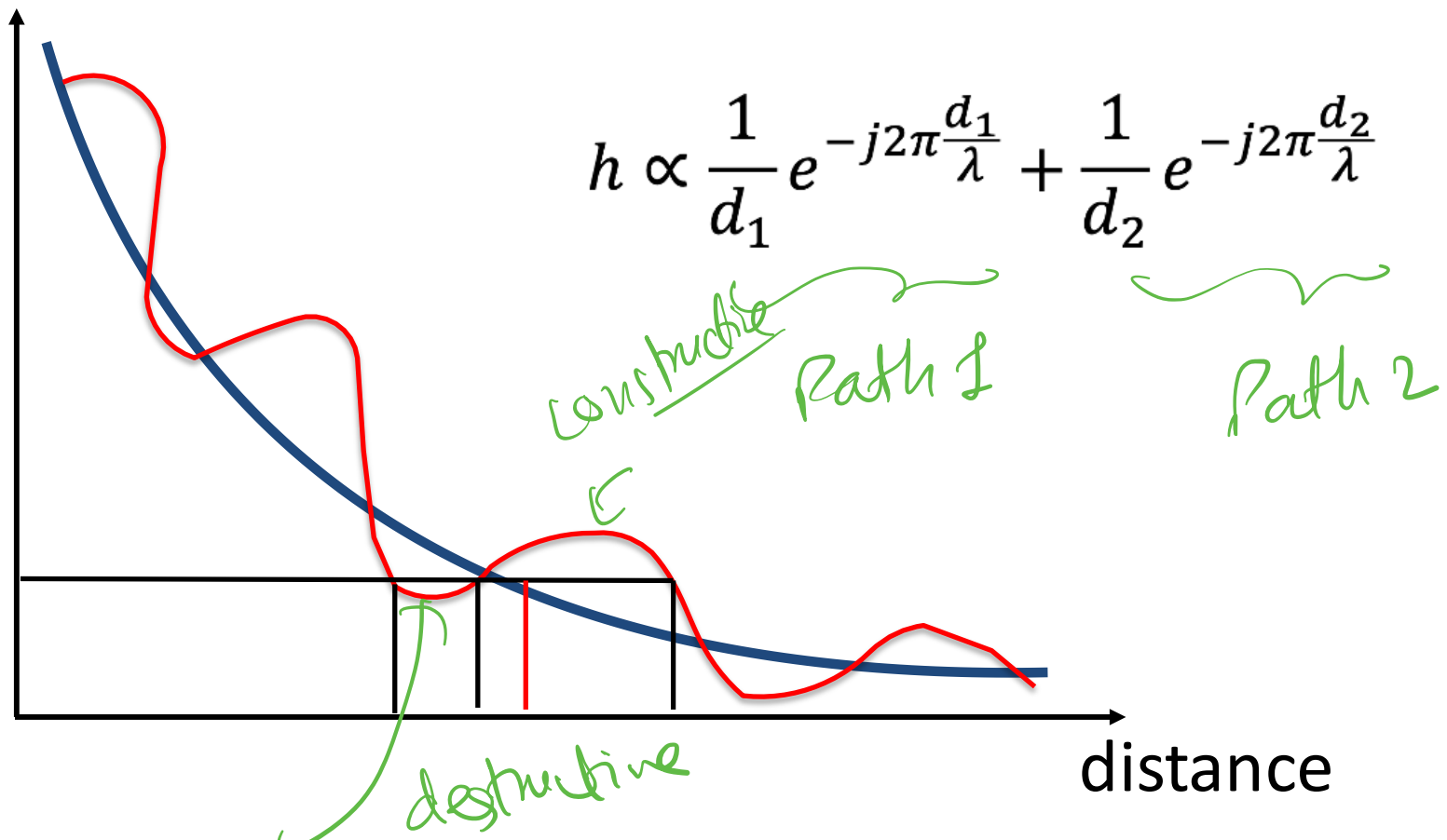
Method 1: RSSI Based Localization

Pros: Very simple, no hardware modifications!

Cons: Highly inaccurate!

Does not work with multipath! or blockage

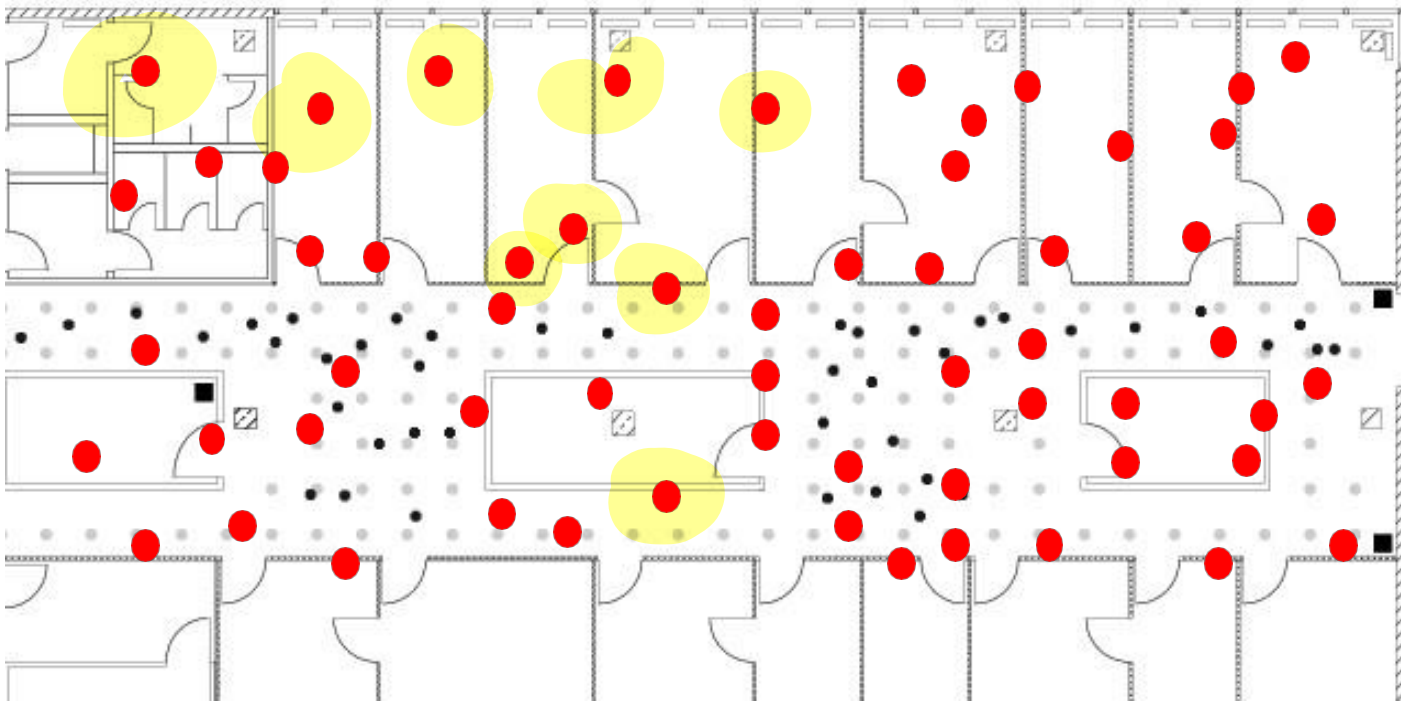
RSSI



Method 1: RSSI Based Localization

Solution: ~~Fingerprinting~~

Measure and records RSSI fingerprints at each location (war-driving)

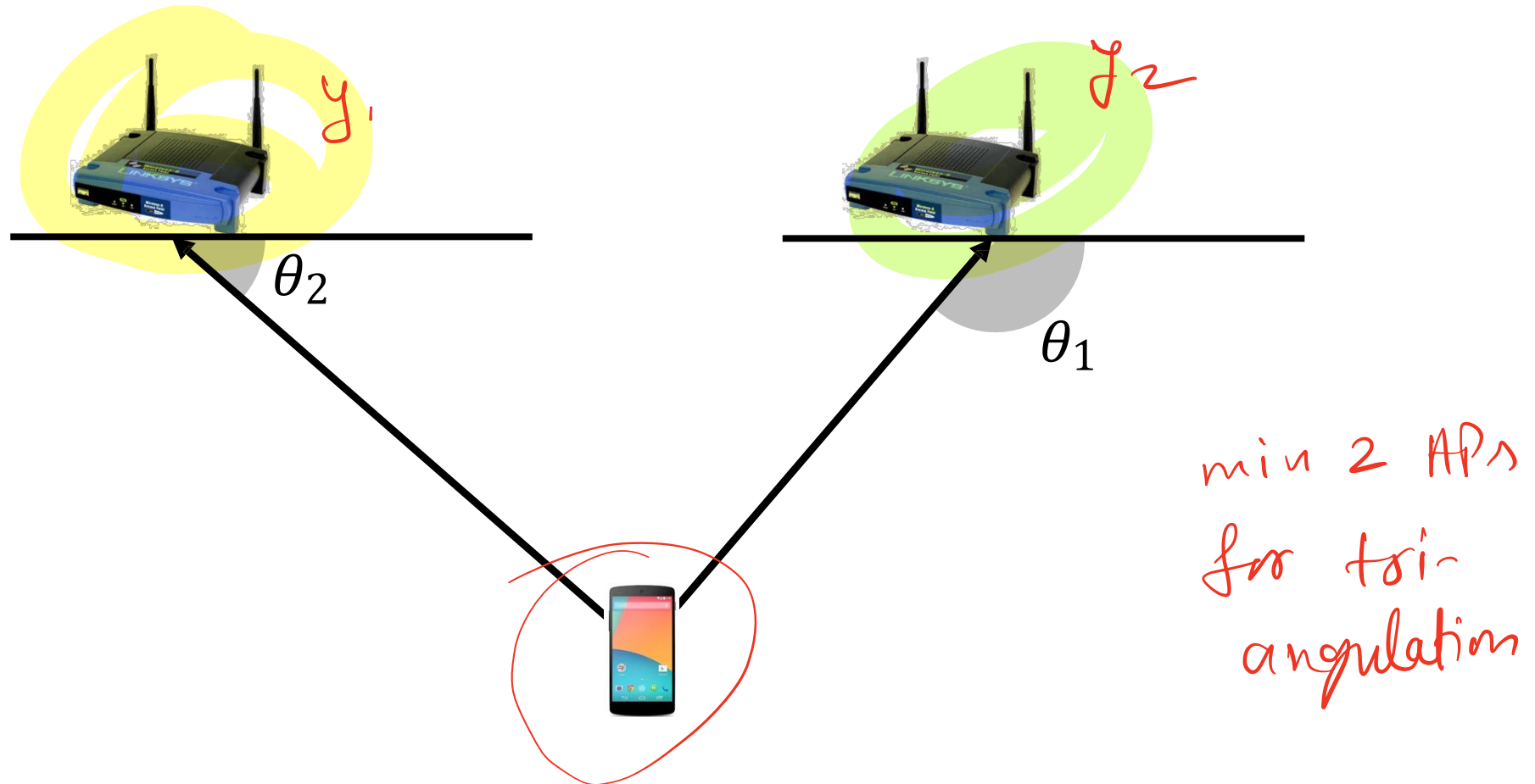


Pros: Works with multipath, No need to know AP locations!

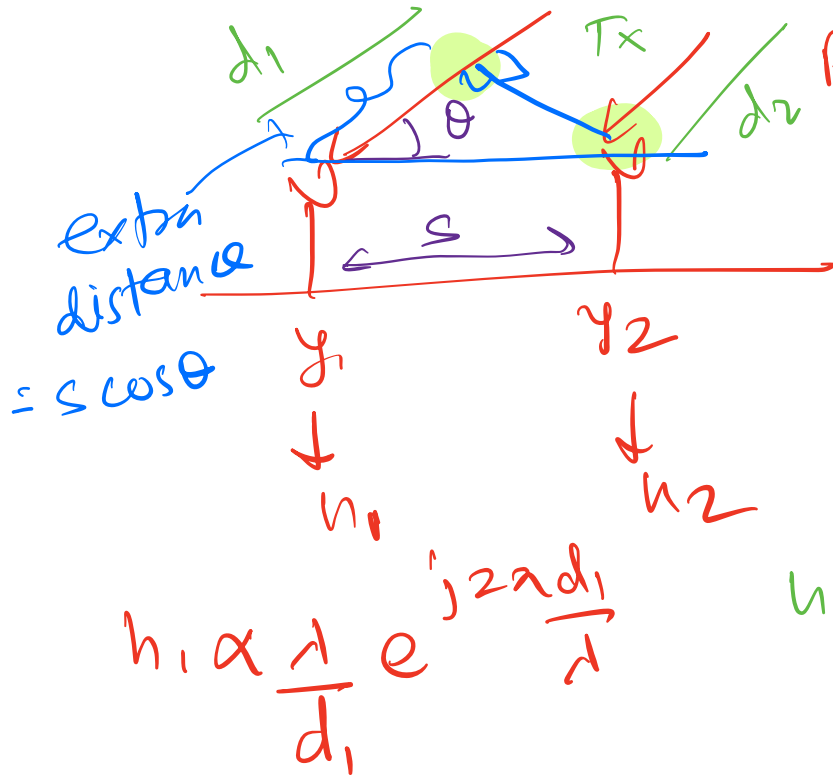
Cons: Changes in environment/movement → change RSSI!
Continuous training is needed. Lots of effort!

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



Triangulation



Parallel rays
(far field assumption)

Assume no multipath

$$h_1 \propto \frac{1}{d_1} e^{j2\pi \frac{d_1}{\lambda}}$$

$$h_2 = \frac{1}{d_2} e^{j2\pi \frac{d_2}{\lambda}}$$

$$d_1 = d_2 - s \cos \theta$$

$$d_2 = d_1 + s \cos \theta$$

$$h_2 = \frac{1}{d_1 + s \cos \theta} e^{j2\pi \frac{d_1}{\lambda} + \frac{s \cos \theta}{\lambda}}$$

$$h_1 = \frac{1}{d_1} e^{j2\pi \frac{d_1}{\lambda}}$$

$$\frac{h_2}{h_1} \propto e^{j2\pi \frac{s \cos \theta}{\lambda}}$$

$\Rightarrow$ estimate $\cos \theta \rightarrow \theta$

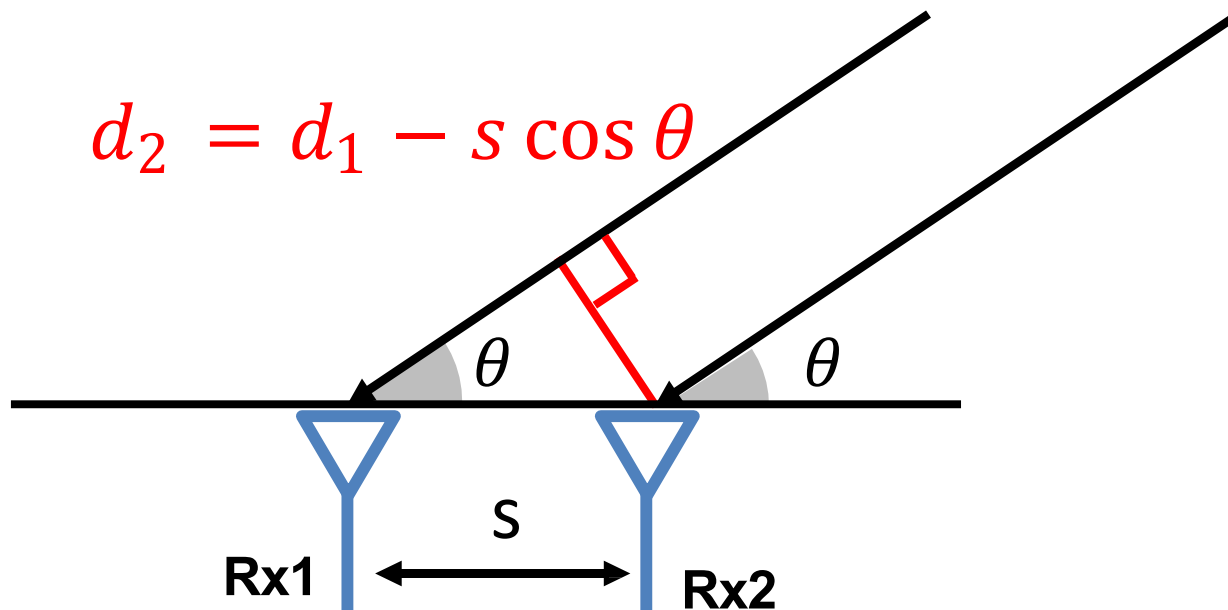
s = distance between antennas

θ = AoA measured from x-axis

Method 2: AoA Based Localization

How to estimate angle

Measure Angle of Arrival (AoA) from device to each AP



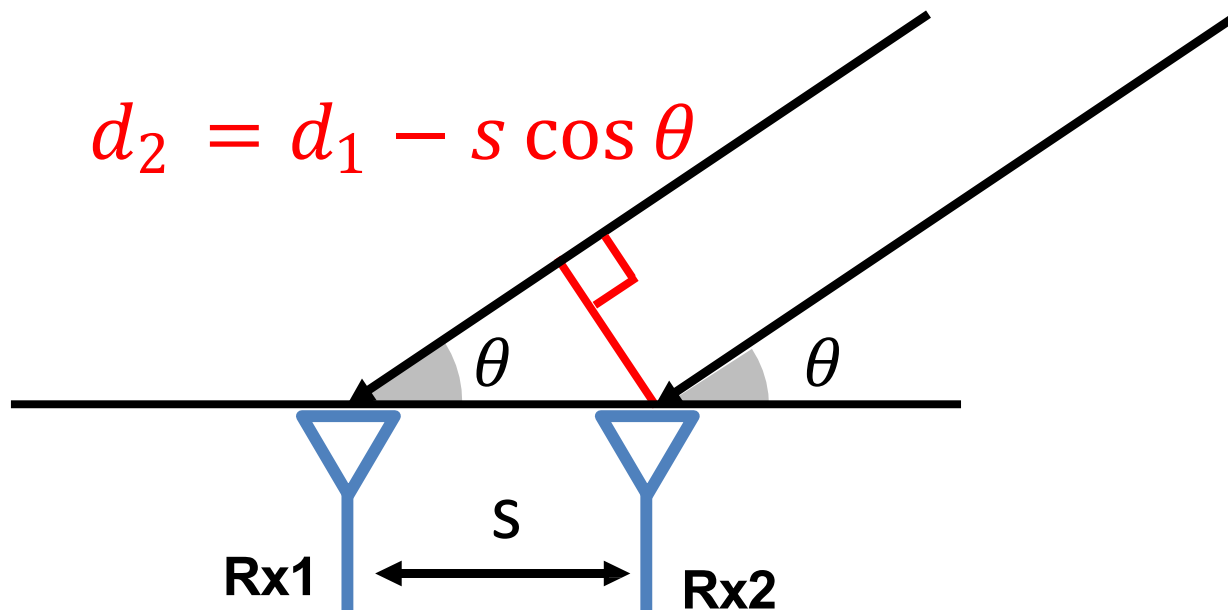
$$d_2 = d_1 - s \cos \theta$$

$$h_1 \propto e^{-j2\pi \frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi \frac{d_2}{\lambda}} = e^{-j2\pi \frac{d_1 - s \cos \theta}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \bmod 2\pi$$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



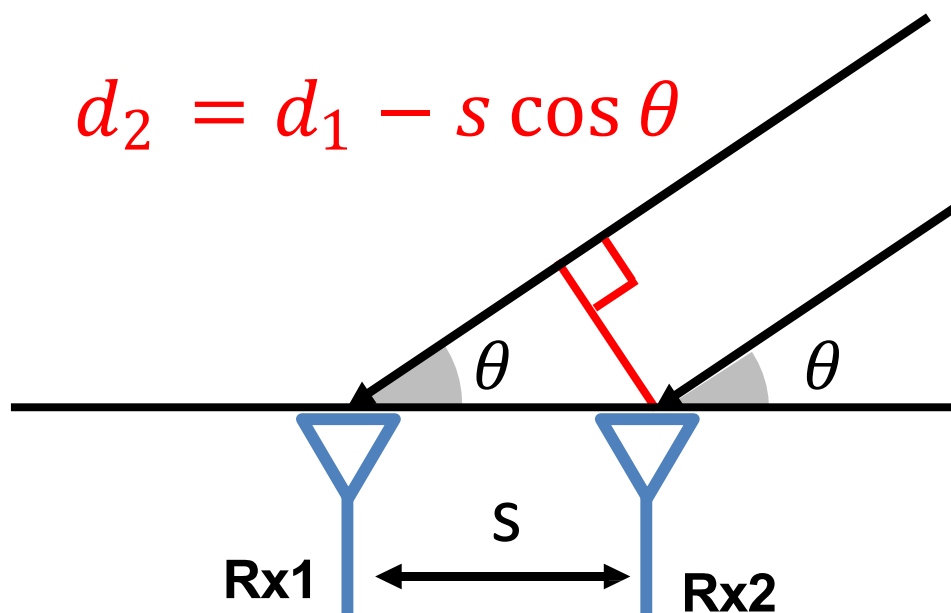
$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \mod 2\pi$$

Ambiguity: $\exists \theta_1 \neq \theta_2 \mid \Delta\Phi_1 = \Delta\Phi_2 \mod 2\pi$

To avoid ambiguity, we want: $-\pi \leq \Delta\Phi \leq \pi$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



We want:

$$-\pi \leq \Delta\Phi \leq \pi$$

We have:

$$-1 \leq \cos \theta \leq 1$$

$$-2\pi \frac{s}{\lambda} \leq 2\pi \frac{s}{\lambda} \cos \theta \leq 2\pi \frac{s}{\lambda}$$

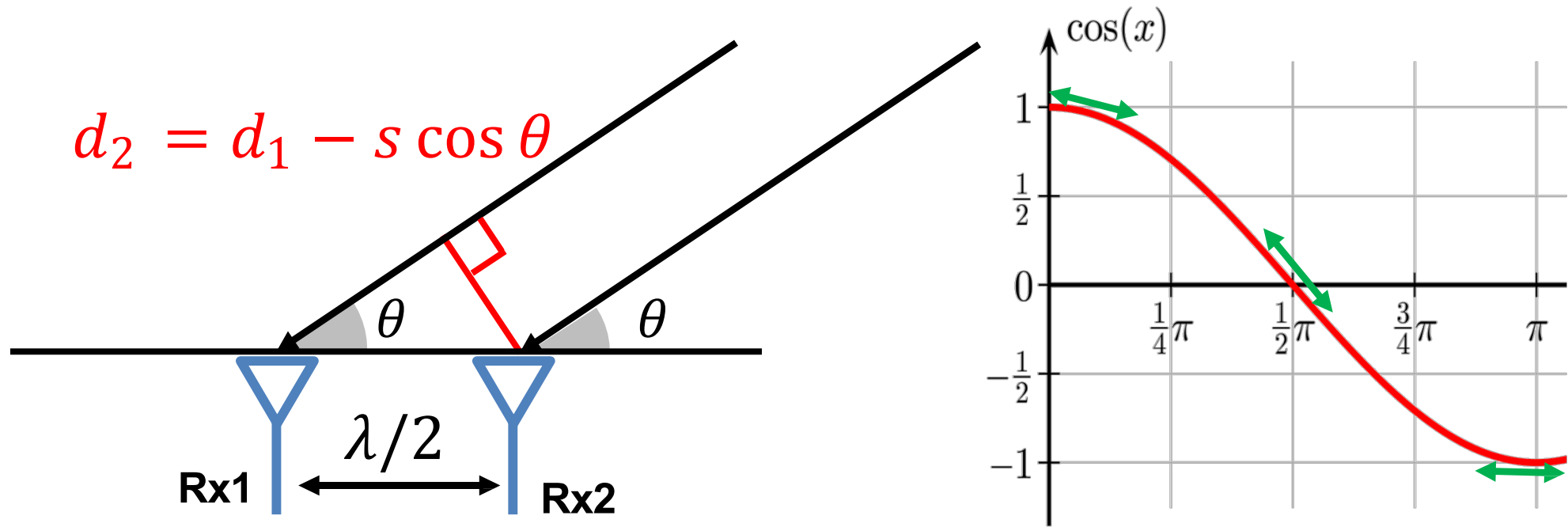
$$-2\pi \frac{s}{\lambda} \leq \Delta\Phi \leq 2\pi \frac{s}{\lambda}$$

$$-\pi \leq \Delta\Phi \leq \pi$$

Set: $s = \lambda/2$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



Pros: More accurate than RSSI, Simple!

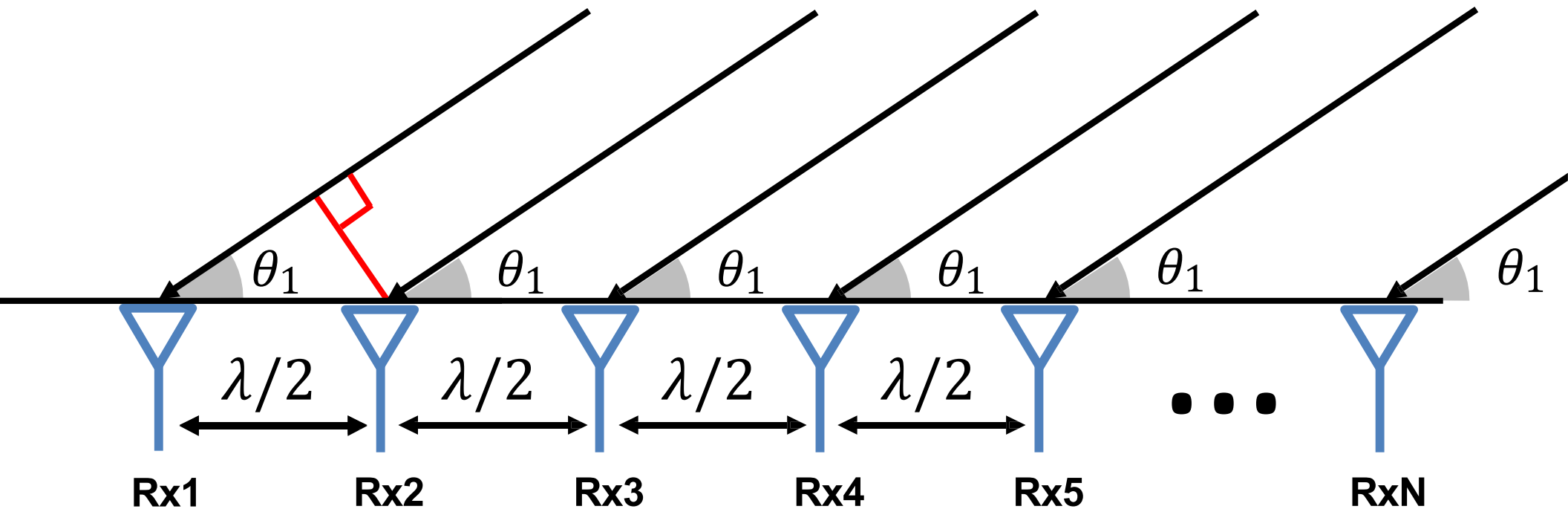
Cons: Ambiguity: $\cos \theta = \cos(-\theta)$

Error not linear with θ due to $\cos \theta$

Requires 2 Antennas separated $\lambda/2$

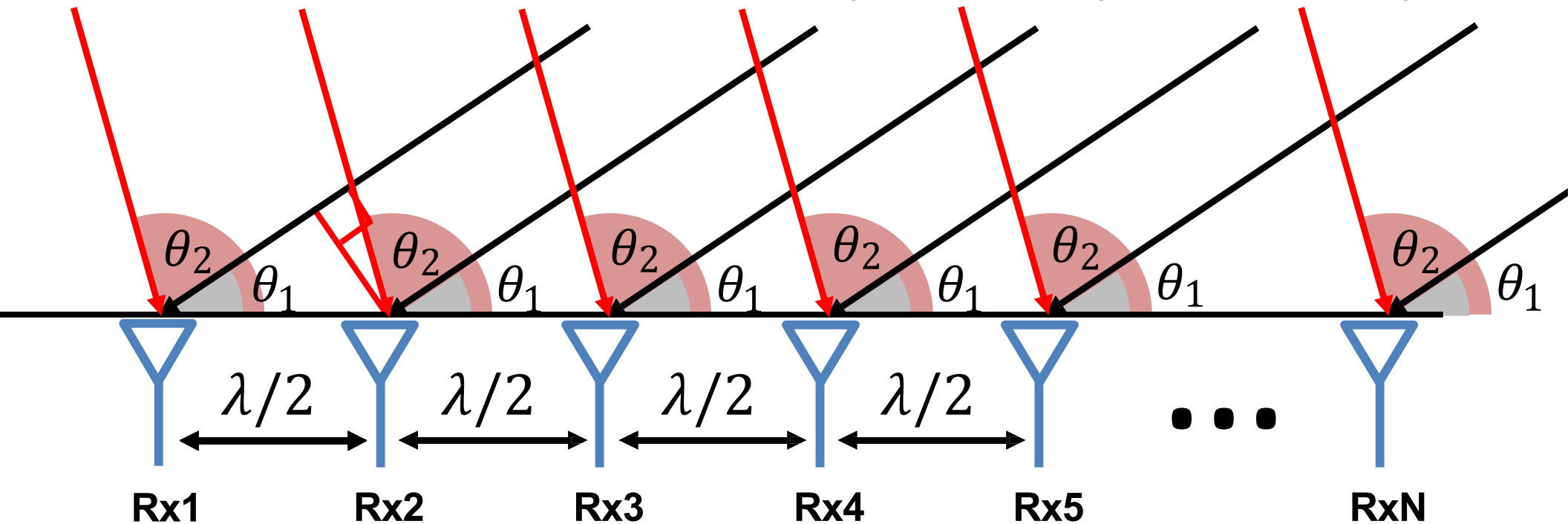
Does not work with multipath!

Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

Method 3: Antenna Arrays (ArrayTrack Paper)



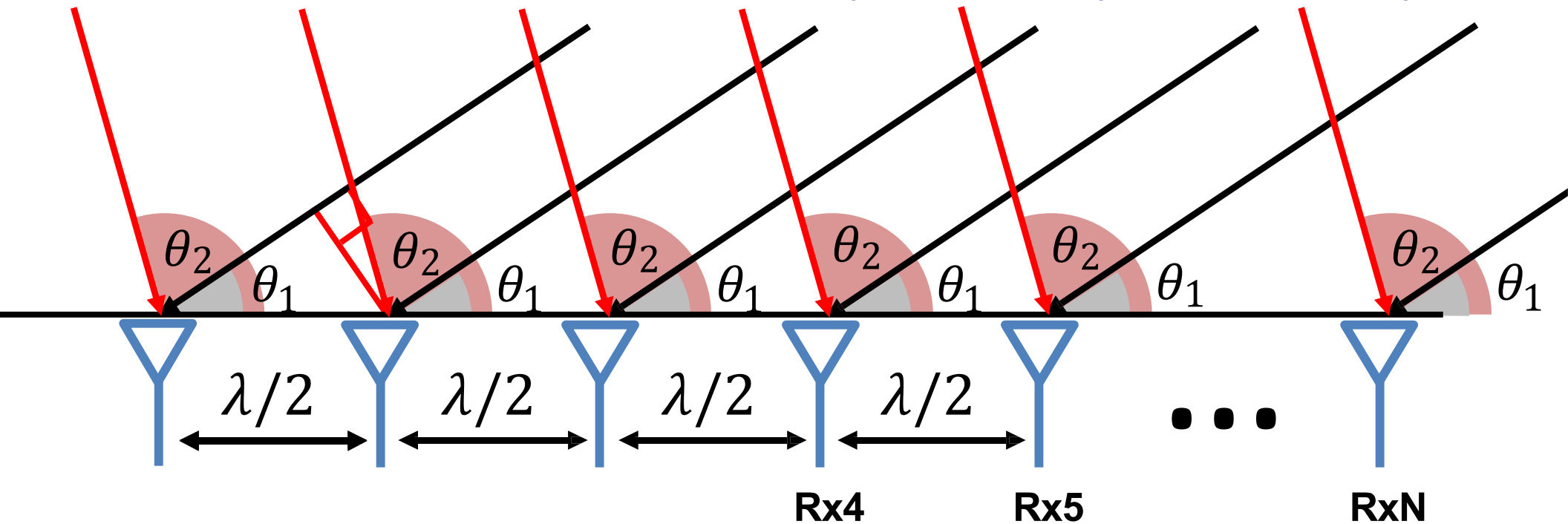
$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

$$+ \alpha_2 e^{-j2\pi \frac{d_2 - k s \cos \theta_2}{\lambda}} + \alpha_2 e^{-j\phi_2 + k\pi \cos \theta_2}$$

For L paths $\rightarrow$

$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Multipath Profile:
$$P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$

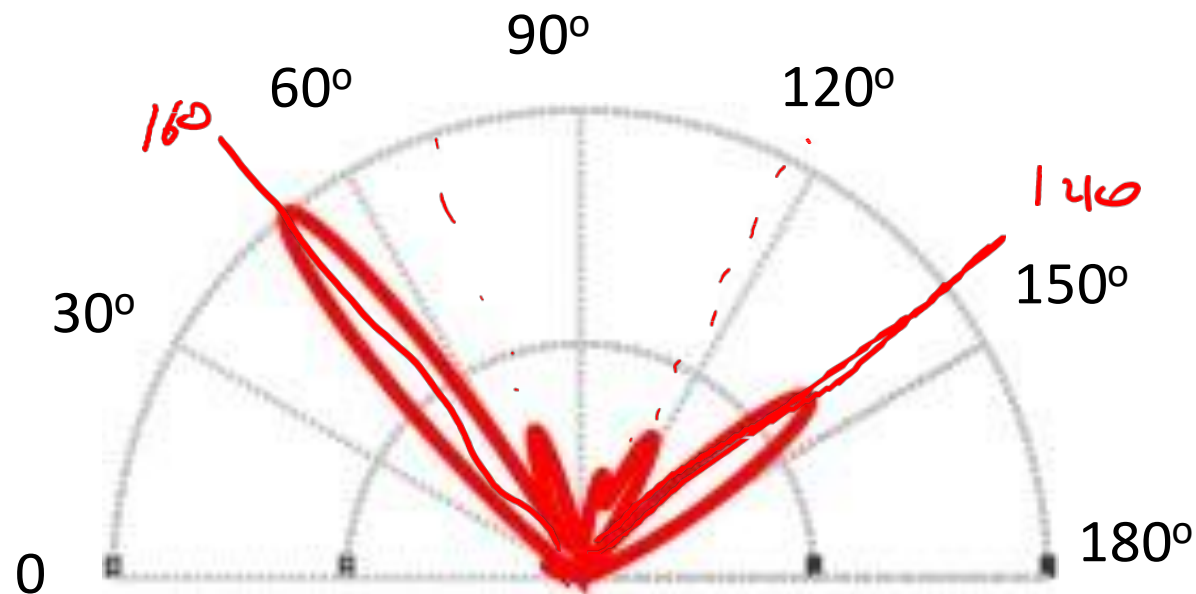
Method 3: Antenna Arrays (ArrayTrack Paper)

$$\begin{aligned}
 P(\theta) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2 & h_k &= \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \\
 P(\theta_1) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \left(\sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \right) e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \alpha_1 e^{-j\phi_1} + \sum_{k=1}^N \left(\sum_{l=2}^L \alpha_l e^{jk\pi (\cos \theta_l - \cos \theta_1) - j\phi_l} \right) \right|^2 \\
 &= \left| N\alpha_1 e^{-j\phi_1} + \sum_{l=2}^L \alpha_l e^{-j\phi_l} \left(\sum_{k=1}^N e^{jk\pi (\cos \theta_l - \cos \theta_1)} \right) \right|^2 \approx N^2 \alpha_1^2
 \end{aligned}$$

Method 3: Antenna Arrays (ArrayTrack Paper)

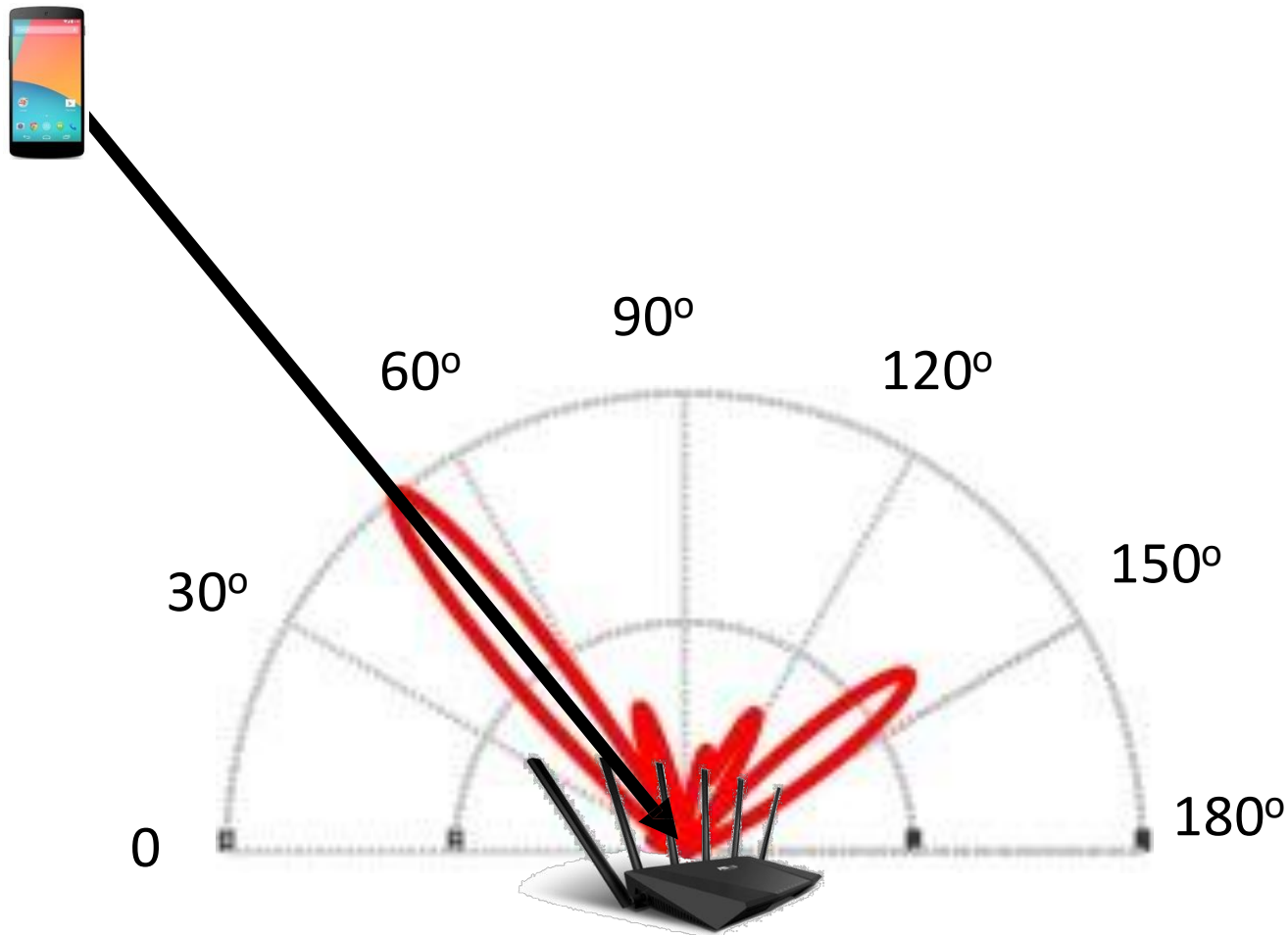
$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

$$\text{Multipath Profile: } P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$



Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?
Strongest Path!

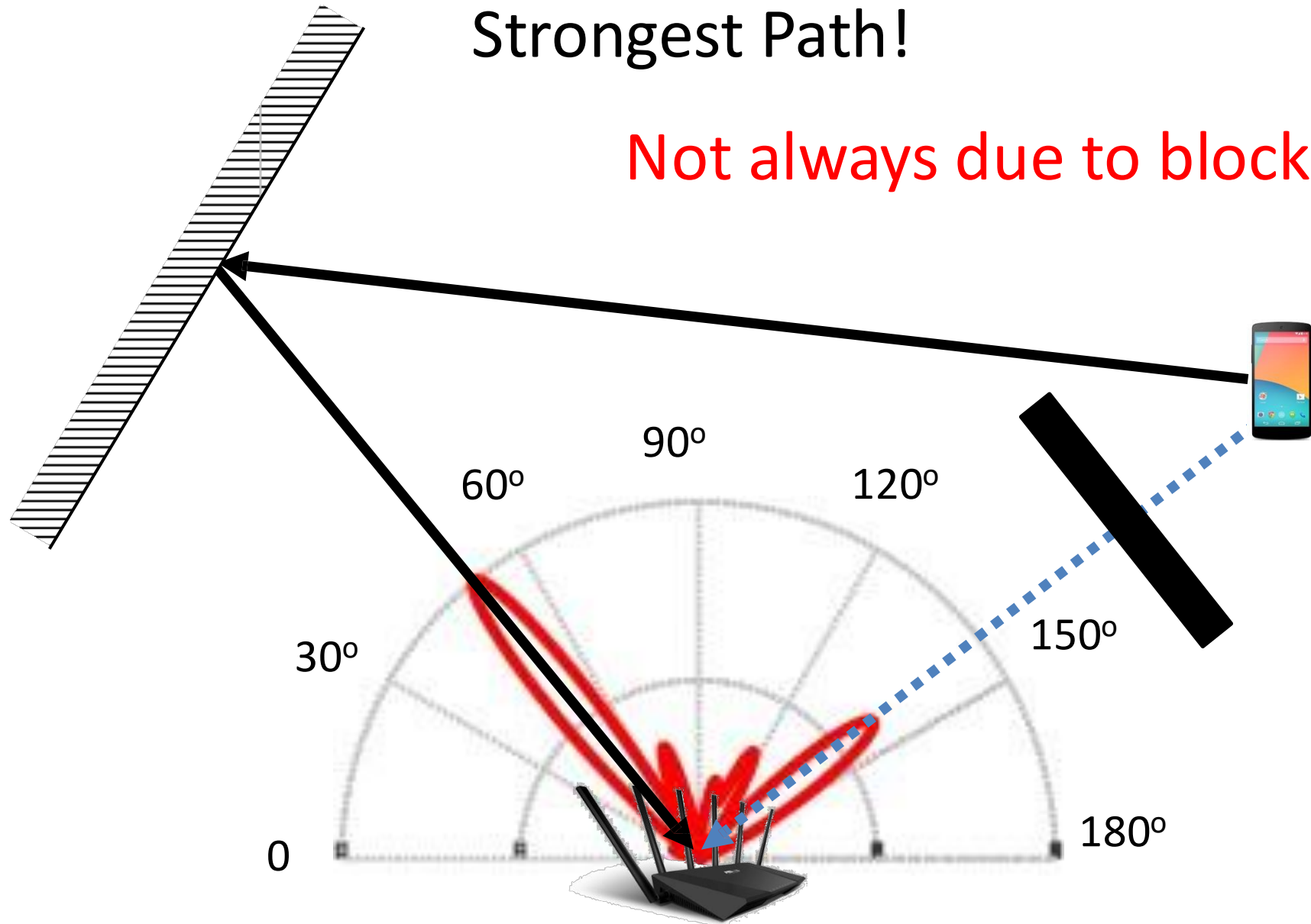


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!

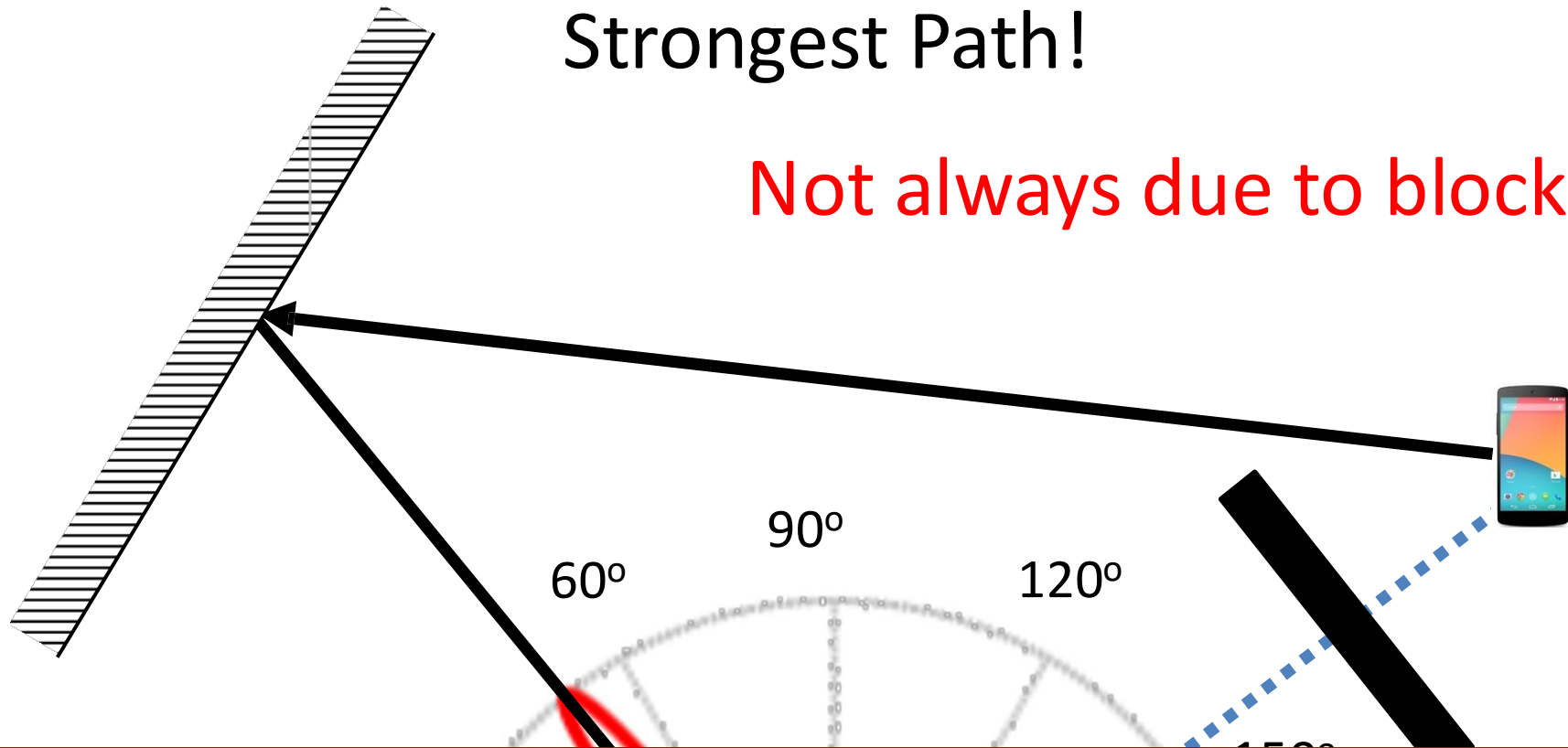


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!



ArrayTrack: Leverage Mobility

- Line of sight path relatively stable with mobility
- Multipath reflection changes faster with mobility

Method 3: Antenna Arrays

Which is the Line-of-Sight Path (Direct Path)?

Shortest Path!

